YINSON PRODUCTION - TREASURY REPORTING

FX Parities Reporting Engine - User Runbook and Update Guide

Runtime questions, expected answers, branching logic, input preparation, troubleshooting, and safe update points

Source script	FX_PARITIES_OVERVIEW_REPORTING_ENGINE-FINAL.py
Prepared for	Operational users, maintainers, and treasury reporting stakeholders
Document purpose	Provide a professional runbook and technical understanding of the FX Parities Reporting Engine

Prepared from the current engine implementation. This document describes the behaviour that is actually present in the supplied script file.

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Guide Purpose

This guide is intended for the person who runs the engine and for the maintainer who needs to keep the user journey understandable.

It explains every visible question the executable can present, what the user should answer, why the answer matters, and what path the engine follows next.

Before You Run the Engine

Before execution, the user should already know whether the report will be generated from the Frankfurter API or from an existing Excel workbook.

If upload mode will be used, the required fields should be checked before launch so the run does not stop at validation or file path issues.

- API mode: internet access required; no user source file is needed.
- Upload mode: a valid Excel path is required; the first sheet must contain DATE, CURRENCY_CODE, and USD_RATE.
- Users should expect three deliverables: one main Excel, one master-data Excel, and one HTML dashboard.

Runtime Questions in Exact Sequence

The engine asks very few questions. That is good for operational simplicity, but it means each answer directly controls an important branch in the process.

Step	Prompt shown	When it appears
Startup confirmation	PLEASE PRESS START TO GENERATE FX REPORT FOR YYYY MM DD	Always, immediately after the opening banner.
Mode selection	DO YOU WANT TO GENERATE THE FX REPORT BY UPLOADING YOUR OWN DATA? (IF NO API CONNECTION WILL BE USED)	Always, after startup confirmation.
Upload file path	PLEASE PROVIDE THE FILE PATH OF THE EXCEL FILE YOU HAVE	Only if the user selected upload mode.
Completion confirmation	PRESS ENTER TO FINISH THE PROCESS	At the end of a successful run.
Error close confirmation	PRESS ENTER TO CLOSE THE PROCESS	Only when an unhandled exception reaches the global error handler.

Operational note: there is also a silent maintenance gate after November 2026. That gate does not show explanatory text, so users must be trained on it in advance.

What the User Should Answer

The following table converts the raw terminal prompts into operational instructions for the user.

Prompt step	What the user should enter	Practical note
Startup confirmation	Press Enter to start the run.	No business data is entered here; it is only a start confirmation gate.
Mode selection	YES / Y / TRUE / 1 to use an existing Excel file, or NO / N / FALSE / 0 to use the API.	Any unrecognised answer causes the script to re-prompt.

Upload file path	A valid path to an existing Excel workbook. The engine reads the first sheet only.	If the path is invalid, the script shows a warning and asks again. Quotes around the path are stripped automatically.
Completion confirmation	Press Enter.	All output files have already been generated and auto-opened before this prompt appears.
Error close confirmation	Press Enter after reviewing the error line shown above it.	KeyboardInterrupt has its own cancellation path and exits with a different message.

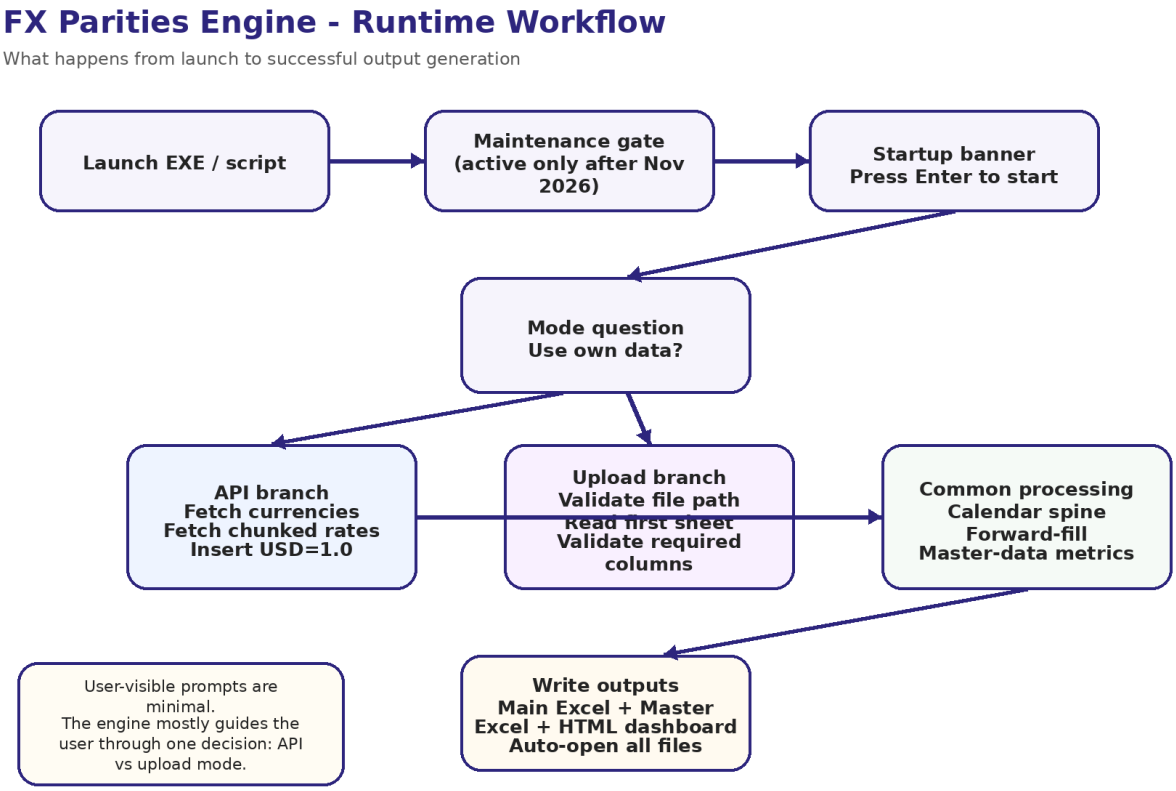
Why Each Question Exists

A user can only respond correctly if they understand the intent of the prompt. The questions are not random. Each one protects a specific control point in the execution flow.

Prompt step	Why the engine asks it
Startup confirmation	Confirms the user is ready to launch the process for the displayed local report date.
Mode selection	This decides the full processing branch: user-upload validation path or automatic Frankfurter API path.
Upload file path	The engine needs a source dataset to normalise, validate, forward-fill, and reshape into the standard output structure.
Completion confirmation	Keeps the terminal window open long enough for the user to review the completion summary when the executable was launched by double-click.
Error close confirmation	Prevents the terminal window from closing instantly, so the user can read the error message.

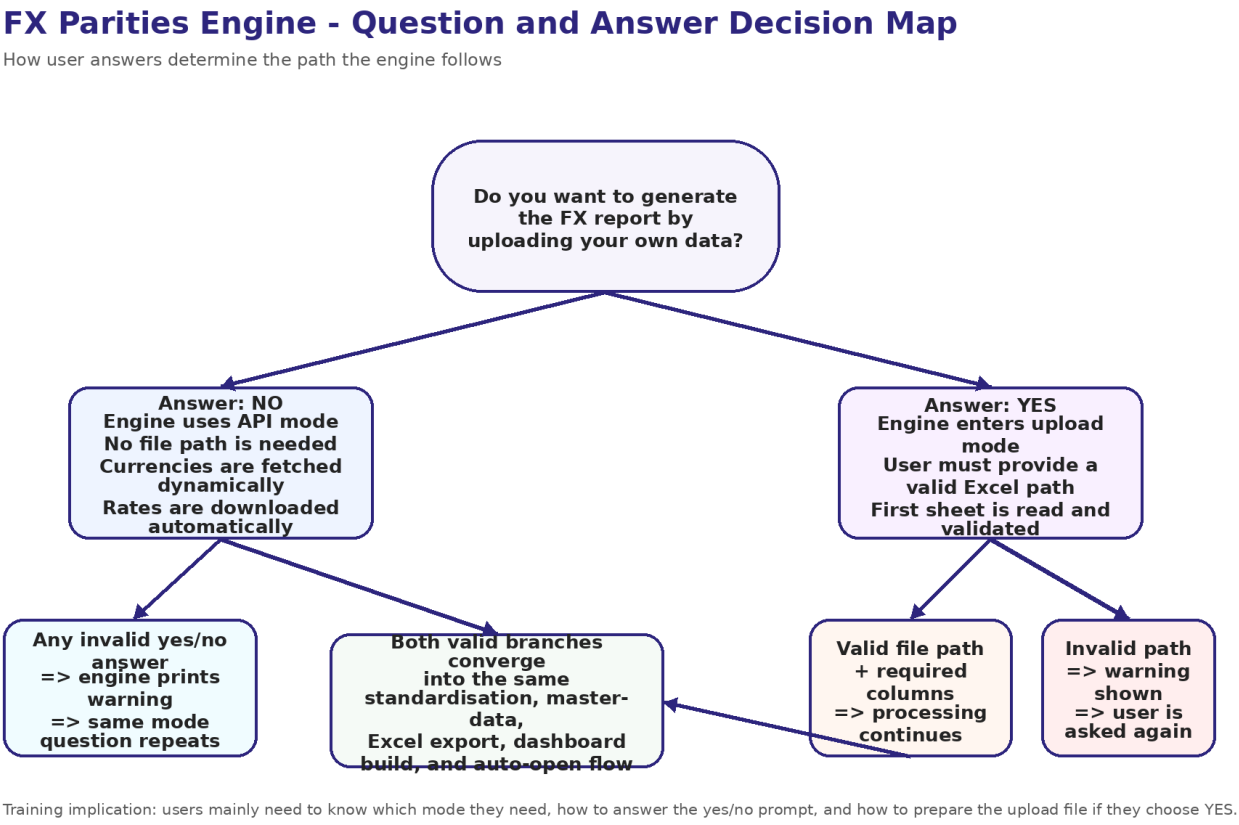
Workflow Visualisation

The diagram below shows the runtime journey from launch to output generation. It is intended to help users and trainers explain what the engine is doing step by step.



Question-to-Path Decision Map

This decision map focuses specifically on how user answers change the route taken by the engine.



Preparing the Upload File

Upload mode is safe only when the file is prepared in the format the engine expects. The user does not need to create every optional field, but the required structure must be respected.

Column	Required?	How the engine treats it
DATE	Required	Report date per row; converted to a date value.
CURRENCY_CODE	Required	ISO currency code; uppercased and trimmed.
USD_RATE	Required	Numeric USD-based FX rate.
PROVIDER	Optional	Defaults to USER_UPLOAD when missing.
IS_FILLED	Optional	Converted to boolean; unrecognised values are treated as False.
LOAD_TIMESTAMP	Optional	Defaults to the current execution timestamp when missing.

- Place the correct dataset on the first worksheet because the engine reads sheet index 0 only.
- Use clean ISO currency codes such as EUR, GBP, SGD, or NOK.
- Ensure USD_RATE is numeric before running the engine.
- Avoid relying on optional metadata fields unless they are intentionally part of the upload process.

What Happens After a Successful Run

After data preparation and enrichment are completed, the engine writes all deliverables to the output directory, opens them automatically, and then shows a final prompt so the user can close the terminal at their own pace.

Deliverable	Purpose
FX_PARITIES_RATES_YYYYMMDD.xlsx	Contains raw daily FX data plus the enriched master-data sheet.
FX_PARITIES_MASTERDATA_YYYYMMDD.xlsx	Contains only the master-data export sheet for downstream use.
FX_PARITIES_DASHBOARD_YYYYMMDD.html	Interactive multi-page FX dashboard with filters, charts, rankings, and heatmaps.

How to Update the Engine Safely

Although this guide is user-facing, maintainers also need a controlled list of the most likely update points. Changes made in these hotspots should be documented and regression-tested because they influence user behaviour, data coverage, or output layout.

Typical change	Where to update	Why it matters
Change API extraction start date	Update START_DATE in the configuration block.	Affects how far back API mode pulls data.
Change output folder or file naming	Update OUTPUT_DIR and/or build_output_paths().	Affects storage location and naming conventions.
Change dashboard focus currencies	Update FOCUS_CURRENCIES.	Affects the Executive Radar page composition.
Change dashboard page grouping	Update OTHER_PAGE_SIZE.	Affects how many currencies appear per “Other Currencies” page.

Change Excel branding colours	Update TAB_COLOR, TITLE_BG, HEADER_BG, BODY_BG, and related style constants.	Affects workbook presentation only.
Change runtime questions	Update print_banner(), choose_data_mode(), and ask_existing_file_path() prompt text.	Affects user interaction and training documentation.
Change upload validation rules	Update load_user_excel_raw().	Affects required columns, default fields, and cleansing behaviour.
Change enrichment metrics	Update build_masterdata().	Affects dashboard measures and the master-data schema.
Make the dashboard truly offline	Replace CDN-based Chart.js / font references inside dashboard_html() with local assets or bundled scripts.	Current template still depends on external CDNs.
Change maintenance validity window	Update _MAINTENANCE_CUTOFF and, if needed, _check_maintenance_gate().	Affects post-cutoff behaviour and compliance control.

Troubleshooting and Common User Mistakes

Most run failures are likely to come from wrong answers, invalid file paths, or incorrectly prepared upload files rather than from complex technical defects.

Scenario	What the user will see
Invalid yes/no answer	The engine prints: INVALID ANSWER. PLEASE TYPE YES OR NO.
Invalid file path	The engine prints: THE FILE PATH PROVIDED COULD NOT BE FOUND. PLEASE TRY AGAIN.
Maintenance gate	After November 2026, the process silently waits up to 10 seconds for Enter. If no acknowledgement is received, it exits immediately.

- If the executable closes instantly after the maintenance validity window, check the cutoff logic and train users on the silent acknowledgement requirement.
- If the user selects upload mode but points to the wrong workbook, the engine will keep asking until a valid path is provided.
- If the upload file exists but the required columns are missing, the run will fail during Excel validation rather than during the initial prompt phase.
- If the dashboard opens but some styling or charts do not load in an isolated environment, review the CDN references in dashboard_html().

Quick Run Checklist

The checklist below can be used as a one-minute operational readiness check before running the engine.

- I know whether I want API mode or upload mode.
- If using upload mode, I already have the correct Excel workbook path.
- The first worksheet contains DATE, CURRENCY_CODE, and USD_RATE.
- I expect three deliverables in the output folder.
- I know the process will auto-open the generated files at the end.
- I know how to respond if the engine asks again because of an invalid yes/no answer or an invalid file path.